

Tutorial Sheet - I

1. Describe all the 2×2 row-reduced echelon matrices.

2. Let $A = \begin{bmatrix} 3 & -1 & 2 \\ 2 & 1 & 1 \\ 1 & -3 & 0 \end{bmatrix}$.

For which $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ does the system $AX=Y$ have a solution?

3. Let A be a $n \times n$ matrix.

(i) Suppose there exists a $n \times n$ matrix B such that $BA=I$. Show that A is invertible.

(ii) Suppose there exists a $n \times n$ matrix C such that $AC=I$. Show that A is invertible.

4. Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 2 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ -4 & 4 \end{bmatrix}$

Find a matrix C such that $CA=B$.
Is the matrix C unique?

5. If A is an $m \times n$ matrix and B is an $n \times m$ matrix with $n < m$, show that AB is not invertible.

6. Let $B \in M_{n \times n}(\mathbb{C})$.

(i) Show that $B' = \{A \in M_{n \times n}(\mathbb{C}) : AB = BA\}$ is a subspace of $M_{n \times n}(\mathbb{C})$.

(ii) Find B' , where $B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & \dots & n \end{bmatrix}_{n \times n}$.

7. Let $X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in M_{n \times 1}(\mathbb{C})$. Consider the

$$\text{Set } LA(X) = \{A \in M_{n \times n}(\mathbb{C}) : AX = 0\}.$$

(i) Show that $LA(X)$ is a subspace of $M_{n \times n}(\mathbb{C})$.

(ii) Show that $LA(X)$ is a left ideal of $M_{n \times n}(\mathbb{C})$, that is $BC \in LA(X)$ whenever $C \in LA(X)$ and $B \in M_{n \times n}(\mathbb{C})$.

(iii) Show that $LA(X) = M_{n \times n}(\mathbb{C})$ if and only if $X = 0$.

8. Let $A, B \in M_{n \times n}(\mathbb{C})$ be such that $AB = 0$.
Give a proof or counter example for each of the following.

(i) $BA = 0$.

(ii) Either $A = 0$ or $B = 0$ (or both).

(iii) If B is invertible then $A = 0$.

(iv) There is a vector $X \neq 0$ such that $BAx = 0$.